

KUPINO, Russian Federation

WMO: 297060

Lat: 54.3869N

Lon: 77.3083E

Elev: 116

StdP: 99.94

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.4	(c) -32.2	(d) -38.4	(e) 0.1	(f) -35.2	(g) -35.5	(h) 0.1	(i) -32.1	(j) 12.4	(k) -10.7	(l) 10.5	(m) -8.4	(n) 2.4	(o) 250	(p) 0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 30.6	(d) 18.6	(e) 28.9	(f) 18.0	(g) 27.1	(h) 17.4	(i) 20.5	(j) 26.6	(k) 19.6	(l) 25.6	(m) 18.8	(n) 24.8	(o) 4.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.5	22.7	17.5	12.7	22.1	16.6	12.0	21.4	59.7	26.3	56.6	25.7	54.1	24.7	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.3	8.2	-37.4	33.8	4.5	2.0	-40.6	35.3	-43.2	36.4	-45.7	37.6	-49.0	39.1
			-37.6	22.1	4.4	1.3	-40.8	23.1	-43.4	23.9	-45.9	24.6	-49.1	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-18.5	-16.1	-6.9	4.8	12.4	18.0	19.6	17.1	11.1	3.5	-6.2	-14.7		
	DBStd	14.68	8.13	7.31	6.02	5.95	5.37	4.19	3.34	3.89	4.74	5.15	7.23	8.00		
	HDD10.0	3856	882	729	524	176	39	2	0	1	42	209	487	765		
	HDD18.3	6079	1141	963	782	406	194	56	26	70	222	460	737	1023		
	CDD10.0	976	0	0	0	20	112	242	297	222	76	8	0	0		
	CDD18.3	157	0	0	0	0	9	45	65	33	5	0	0	0		
	CDH23.3	1652	0	0	0	8	168	456	577	368	75	0	0	0		
	CDH26.7	458	0	0	0	1	39	128	164	109	18	0	0	0		
(14) Wind	WSAvg	3.9	4.0	4.2	4.5	4.4	4.2	3.4	3.2	3.3	3.5	3.9	4.3	4.3		
(15) Precipitation	PrecAvg	311	12	9	9	15	25	39	54	53	31	25	19	14		
	PrecMax	464	38	27	34	49	68	102	127	145	87	64	53	36		
	PrecMin	186	2	1	1	0	4	4	0	6	0	2	1	2		
	PrecStd	60	8	6	6	13	15	25	32	33	19	16	12	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	0.9	6.7	24.4	30.5	32.2	33.0	32.7	29.1	20.3	8.1	1.3		
		MCWB	0.1	0.0	3.4	13.0	16.4	18.3	20.1	19.0	15.8	11.1	5.7	0.3		
	2%	DB	-1.9	-0.8	3.8	20.5	27.5	29.9	30.3	29.6	25.4	16.8	6.0	0.2		
		MCWB	-2.7	-1.5	1.9	11.6	15.3	18.2	19.4	18.4	14.8	9.8	4.2	-0.3		
	5%	DB	-4.1	-2.9	2.3	17.4	25.3	28.0	28.5	27.2	22.1	13.6	3.9	-1.6		
		MCWB	-4.7	-3.6	0.9	9.9	14.2	17.9	18.8	17.4	13.5	8.5	2.6	-2.3		
	10%	DB	-6.2	-5.1	1.1	14.3	22.4	25.9	26.6	25.1	19.3	10.8	2.1	-3.7		
		MCWB	-6.8	-5.8	0.0	8.5	13.0	17.1	18.5	16.9	12.2	7.0	0.9	-4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.1	3.7	14.1	18.2	21.2	22.0	20.8	17.7	12.1	6.6	0.6		
		MCDB	0.7	0.7	6.0	22.1	27.8	28.2	29.0	27.2	26.4	17.3	7.4	1.0		
	2%	WB	-2.6	-1.5	2.1	12.4	16.2	19.6	20.9	19.6	15.6	10.6	4.4	-0.3		
		MCDB	-2.1	-0.9	3.5	19.1	24.9	26.2	26.8	26.3	22.9	15.5	5.8	0.2		
	5%	WB	-4.7	-3.5	1.1	10.4	14.9	18.8	20.0	18.6	14.2	8.9	2.7	-2.2		
		MCDB	-4.1	-2.9	2.2	16.5	23.3	25.3	25.5	25.0	21.1	12.8	4.0	-1.6		
	10%	WB	-6.8	-5.8	0.1	8.7	13.7	17.9	19.3	17.6	12.8	7.4	1.0	-4.3		
		MCDB	-6.2	-5.2	1.1	14.1	21.0	24.1	24.7	23.8	18.3	10.6	1.9	-3.7		
(35) Mean Daily Temperature Range	MDBR		7.9	8.3	8.3	10.0	12.1	11.1	10.5	11.3	10.6	8.4	6.6	7.7		
	5% DB	MCDBR	8.4	7.6	7.0	15.0	16.1	13.9	13.6	14.9	15.6	13.2	6.3	8.0		
		MCWBR	8.4	7.5	5.9	7.9	6.9	5.6	5.1	5.8	7.2	7.4	5.2	7.9		
	5% WB	MCDBR	8.5	7.6	6.6	14.0	14.0	11.7	11.1	12.4	13.8	11.8	6.1	7.8		
		MCWBR	8.5	7.5	5.8	7.7	7.0	5.5	4.9	5.7	7.1	7.1	5.3	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.321	0.379	0.403	0.404	0.411	0.435	0.421	0.368	0.351	0.325	0.292		
	taud		2.033	1.993	1.932	2.157	2.224	2.291	2.223	2.228	2.330	2.250	2.094	1.990		
	Ebn at Noon		615	729	766	817	840	838	807	793	792	703	563	508		
	Edh at Noon		72	106	143	132	130	123	130	122	96	83	68	61		
(44) All-Sky Solar Radiation	RadAvg		1.05	2.15	3.75	4.71	5.73	6.11	5.57	4.61	3.29	1.69	0.80	0.69		
	RadStd		0.07	0.09	0.25	0.35	0.47	0.44	0.37	0.38	0.32	0.22	0.11	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page